

Day 05 - Linux Troubleshooting skill;

⇒ CPU, Memory, and I/O

Runbook:-

* Environment basics:-

Command: `uname -a`

This Command: `uname -a` : `uname` :- Unix Name (show only kernel)
-a: all information about kernel :-

Example:-

Linux Matrix 5.15.162.4-microsoft-wsl2 #1 SMP
→ The Nov 5 00:21:55 UTC 2024
x86_64 x86_64 x86_64 GNU/Linux

Linux → kernel / operating system name
Matrix → hostname of the machine
5.15.162 wsl2 → Linux kernel version running on the system

#1 → Build number of the kernel
SMP → System Symmetric Multi-processing
[Support multiple CPU cores]

The → Day kernel was built
Nov → Month kernel was built
5 → Date kernel was built
00:21:55 → Time kernel was built
UTC → Time zone used during build
year kernel was built

x86_64

x86_64

x86_64

GNU/Linux

CPU architecture

Machine hardware platform

Processor type

Operating system kernel

+ kernel combination

2. lsb_release -a: [LSB Linux Standard Base]
 → display detailed Linux distribution information.

Exmp:-

Distributor ID: Ubuntu

Description: Ubuntu 24.04.04 LTS

Release: 24.04

Codename: noble

LTS: Long Term Support

24.04: [24 year, 04 (4 month)]

→ 24 April Release

3. hostnmectl

→ view and configure the system hostname and related system information [OS kernel, architecture]

* file system security :-

4. mkdir /tmp/junkbox-demo

This command create a directory - junkbox-demo inside /tmp (directory)

5. cp /etc/hosts /tmp/junkbox-demo/hosts-copy

This command copy the hosts file information to new file hosts-copy inside the /tmp/junkbox-demo destination path: /tmp/junkbox-demo/hosts-copy

6. ls -l /tmp/xx-back demo

This command verify your back-copy is available or not after copy back (file copy to) the /tmp/xx-back-demo/back-copy. [as backup]

[One large service/process]

I choose the hpx service : systemctl status hpx
→ CPU/Memory :-

7. top :-

Realtime-System monitoring
realtime → CPU, mem, memory, running processes

top -p (process id)

→ check the a particular process with id

8. ps -o pid, pcpu, pmem, comm -p <pid>

This command use to show a specific process information about the pid, pcpu, pmem, comm (process) -p (process id)

but it is not real time it is average value show

→ real time you use : top -p (process id)

and -

ps -C hpx -o pid, ppid, %cpu, %mem, cmd

9. `ps aux -sort --cpu 1200`

This command use to show sort CPU
Usage first and which process use the CPU

10. `df -h 1 df -l`

This command check any free space
disk usage:

11. `free -h`

show the free memory information
in human readable

* Disk I/O

12. `df -h`

disk free [human readable]

This command show disk utilization

13. `du -sh <file path> or <directory path>`

~~du~~ : disk usage

du : disk usage

-sh : ~~summary~~ human readable

This command use to check the
which file use or directory how
many space use in the disk.

→ ~~du~~ ~~du~~ -l /var/log/pass --max-depth
2

14. **iostat**:- Input/output statistics
CPU utilization
disk I/O performance
Read/write activity of storage devices

15. **iostat -x 1**

-x:- extended statistics report

1:- set the interval time to 1 second.

This command shows CPU utility and storage device statistics with more information every 1 second in real time.

16. **vmstat**:- virtual memory statistics

This command shows the process, memory swap, disk I/O, system activity, CPU usage.

It gives a quick health snapshot of the system in a single command.

vmstat 1 [give me every 1 ~~one~~ one second]

vmstat 1 5 [every 1 second 5 times]

vmstat -s [summary statistics]

* Network

12 SS -ulph

This Command is used to check
which service is use which port

→ also use this

→ ss -tudo ss -ulph | grep 80
↳ This Command is use to check
port so which process use.

❖ curl -I <service-endpoint>/ping
-s: fetch only header

This Command use to test / verify,
whether and service to check the
service response to the machine or
not.

This

/ping endpoint

This Command is health check URL
that test what 200 OK and confirms
the service is running!

→ curl http://localhost:8080/ping
output:

→ pong

This message show your
server is running or alive!

* Logs:-

19. journalctl -h <service> -h 50

This command show the service of the log with how many line of log would you see:-

Example:-

journalctl -h nginx -h 50

20: journalctl -h nginx -f

→ This command shows log in real time.
eg:-

--since (today) → This show today

21:- tail -t 50 /var

tail -h 50 /var/log/nginx/access.log

tail -h 50 /var/log/nginx/error.log

This command shows the last 50 lines of log file whether the file is access.log or error.log.

tail -f /var/log/nginx/access.log
Real time log view the log file